

My first markdown cell in Jupyter

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
import warnings
warnings.filterwarnings('ignore')

print("✅ All libraries imported successfully!")
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
```

✅ All libraries imported successfully!
Pandas version: 2.2.3
NumPy version: 2.1.3

```
from sklearn.datasets import load_iris

iris = load_iris()
print("Dataset loaded successfully!")
print(f"Dataset shape: {iris.data.shape}")
print(f"Features: {iris.feature_names}")
print(f"Target classes: {iris.target_names}")
```

Dataset loaded successfully!
Dataset shape: (150, 4)
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Target classes: ['setosa' 'versicolor' 'virginica']

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target_names[iris.target]

print("First 5 rows of our dataset:")
print(df.head())
print("\nDataset info:")
print(df.info())
```

First 5 rows of our dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

```
species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

Dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	sepal length (cm)	150 non-null	float64
1	sepal width (cm)	150 non-null	float64
2	petal length (cm)	150 non-null	float64
3	petal width (cm)	150 non-null	float64
4	species	150 non-null	object

dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None

```
print("Basic Statistics for Iris Dataset:")
print("=" * 40)
print("\nMean values by species:")
print(df.groupby('species').mean())
print("\nSamples per species:")
print(df['species'].value_counts())
```

Basic Statistics for Iris Dataset:

=====

Mean values by species:

	sepal length (cm)	sepal width (cm)	petal length (cm)	\
species				
setosa	5.006	3.428	1.462	
versicolor	5.936	2.770	4.260	
virginica	6.588	2.974	5.552	

	petal width (cm)
species	
setosa	0.246
versicolor	1.326
virginica	2.026

Samples per species:

species	
setosa	50
versicolor	50
virginica	50

Name: count, dtype: int64

```
sepal_lengths = df['sepal length (cm)']
mean_sepal_length = np.mean(sepal_lengths)
std_sepal_length = np.std(sepal_lengths)

print(f"Mean sepal length: {mean_sepal_length:.2f} cm")
print(f"Standard deviation: {std_sepal_length:.2f} cm")

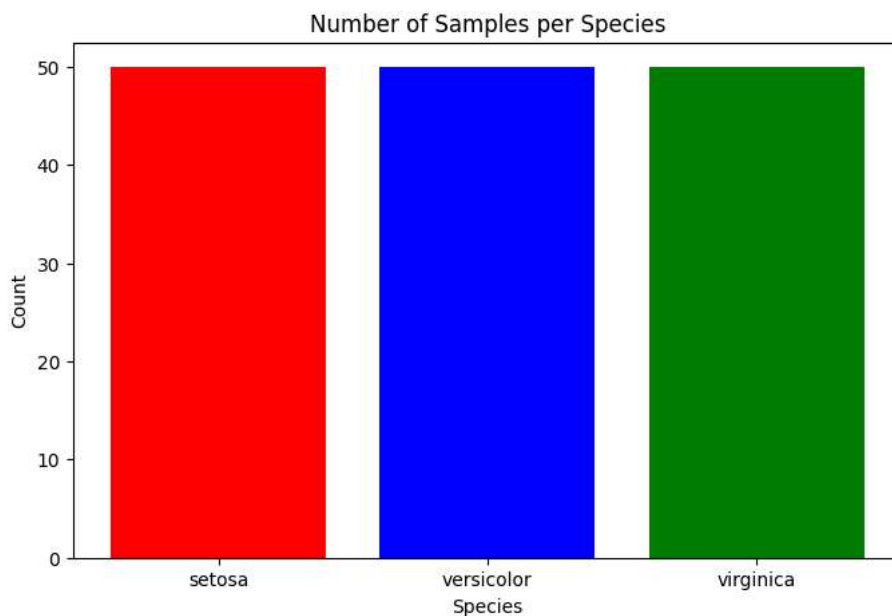
assert isinstance(mean_sepal_length, (float, np.floating)), "Mean should be a number"
assert isinstance(std_sepal_length, (float, np.floating)), "Std should be a number"
print("✅ Task 1 completed successfully!")
```

Mean sepal length: 5.84 cm
 Standard deviation: 0.83 cm
 ✅ Task 1 completed successfully!

```
species_counts = df['species'].value_counts()

plt.figure(figsize=(8, 5))
plt.bar(species_counts.index, species_counts.values, color=['red', 'blue', 'green'])
plt.title('Number of Samples per Species')
plt.xlabel('Species')
plt.ylabel('Count')
plt.show()

print(f"Species distribution: {dict(species_counts)}")
print("✅ Task 2 completed successfully!")
```



Species distribution: {'setosa': np.int64(50), 'versicolor': np.int64(50), 'virginica': np.int64(50)}
 ✅ Task 2 completed successfully!

